

MBA Fastrack 2025 (CAT + OMETs)

QUANTITATIVE APTITUDE

DPP: 5

Inequalities

Q1 For how many natural number values of x will $(x - 2)(x + 3)$ be positive?

- (A) 0 (B) 1
(C) 2 (D) Infinite

Q2 Let 'N' be a natural number. How many values of 'N' satisfy $2N + 5 > 3N$?

- (A) 5 (B) 4
(C) 3 (D) 2

Q3 Find the range of x in $(x - 3)(x - 2) > 30$.

- (A) $x < 2$ or $x > 3$
(B) (2, 3)
(C) (-3, 8)
(D) $x < -3$ or $x > 8$

Q4 If $15x + 19 \leq 21x + 4$ and $17x + 18 \geq 11x - 12$, find the solution for the given inequalities.

- (A) $[\frac{5}{2}, \infty)$
(B) $(-\frac{5}{2}, \infty)$
(C) $[-\frac{5}{2}, \frac{5}{2}]$
(D) None of the above

Q5 Solve the quadratic inequality $x^2 - 15x + 56 \leq 0$

- (A) (7, 8) (B) [7, 8]
(C) [7, 8] (D) [7, 8]

Q6 If $x \geq 2$ and $y \leq 0$, then the greatest possible value of $(y - 3x)$ is =

- (A) -6 (B) -7
(C) -5 (D) 0

Q7 $(x-2)(x-4)(x-8) \leq 0$, where x is a positive integer. If a and b respectively represent the minimum and maximum possible values of $(x + 2)$, then $(a + b) = ?$

- (A) 10 (B) 11
(C) 12 (D) 13

Q8 If x and y are integers, then what is the maximum possible value of x that satisfies the inequality $3x + 5y < 7$?

- (A) 0
(B) 1
(C) 2
(D) Cannot be determined

Q9 How many integral values of x satisfy the inequality $x^2 - 7x + 12 \leq 0$

Q10 If $(x-2^2)(x-2^4)(x-2^6) < 0$ and $(x^2 - 3x - 18) \leq 0$, then find the number of integral values that x can assume.

- (A) 10 (B) 9
(C) 8 (D) 7

Q11 How many positive integer values can x take that satisfy the inequality $(x-8)(x-10)(x-12)(x-100) < 0$?

Q12 If $(x^2 - 2x - 8)(x^2 - x - 12) \leq 0$, then how many natural number values can be assumed by x ?

- (A) 0 (B) 1
(C) 2 (D) 3

Q13 For how many integral values of x does the inequality $(x-3)(x-6)(x-9)\dots(x-108) \leq 0$ hold true?

- (A) 36 (B) 72
(C) 75 (D) 105

Q14 If $x^2 - 11x + 24 \leq 0$, then for how many positive integer values of x , $3 \geq \frac{7x+24}{4x+1}$?



Q15 If $\frac{1}{x+8} \leq \frac{1}{3x-5}$, find the number of positive integer values of x that satisfy the inequality.

Q16 If $(x-1)^{2017}(x-3)^{2019}(x-5)^{2021}(x-7)^{2023} < 0$, then how many integral values $3x$ can assume?

Q17 If $\frac{x^2-6}{2x+7} \leq 3$, then how many prime number values can x have?

Q18 Find the smallest natural number ' x ' that satisfies the following inequality:

$$\frac{x^3-8x^2+20x-16}{(x^2-x-30)} \leq 0$$

Q19 How many integral values x can assume where

$$\frac{(x-10)(x-30)(x-50)}{(x-20)(x-40)(x-60)} \leq 0$$

- (A) 10 (B) 20
(C) 30 (D) 33

Q20 Solve the following inequality:

$$x^3 - 37x - 84 \geq 0$$

- (A) $x \in (-4, -3) \cup (7, \infty)$
(B) $x \in (-\infty, -4) \cup (-3, 7)$
(C) $x \in (-4, -3) \cup [3, 7]$
(D) $x \in [-4, -3] \cup [7, \infty)$

Q21 Find the range of y if $y^6 - 37y^3 - 1728 \leq 0$

- (A) $[-27, 64]$ (B) $[3, 8]$
(C) $[-8, 8]$ (D) $[-3, 4]$

Q22 If $\frac{(x-2)^{11}(x-8)^{13}}{(x-10)^9(x-6)^7} \leq 0$, then the number of integer values of x that satisfy the inequality is:

Q23 Find the sum of all the integral solutions of the following inequality.

$$\frac{6x^2-25x+11}{x^2+4x+5} < 0$$

- (A) 5 (B) 6
(C) 7 (D) 8

Q24 The number of possible integer values satisfying the following inequality is

$$\frac{x^3-14x^2+59x-70}{(x-1)(x-4)^2} \leq 0$$

- (A) 5 (B) 4
(C) 3 (D) 2

Q25 Within the interval $x \in [0, 29)$, how many integer values of x satisfy the following inequality?

$$\frac{x+9}{x-3} + \frac{1}{x-5} \geq \frac{x^2-6x-26}{x^2-8x+15}$$

Q26 How many integer values of x satisfy the below inequality: $\frac{7}{5} < \frac{x}{x+4} < \frac{9}{5}$

- (A) 3 (B) 4
(C) 5 (D) 6

Q27 If x is real, the expression $\frac{-x^2+6x-1}{x}$ takes all values which do not lie in the range $[a, b]$, then a and b are respectively:

- (A) 4, 8 (B) 3, 7
(C) 4, 7 (D) None of these

Q28 Find the sum of all the distinct integer values of x that satisfy both the following inequalities:

$$(5+x)^4(2-x)^3 < 0 \text{ and } -(5-x)^3(9+x)^5 < 0.$$

Q29 The absolute difference between the largest and smallest possible integer values of x satisfying the following inequalities is:

$$\frac{1}{x^2+x-12} - \frac{1}{x^2+11x+28} \leq 0, -9 \leq x \leq 9$$

- (A) 9 (B) 10
(C) 11 (D) 12

Q30 If $6 \geq \frac{x+4}{2x+1} \geq 3$, then the number of integral values assumed by $55x$ will be _____.

- (A) 20 (B) 21
(C) 22 (D) 23



Answer Key

Q1 D
Q2 B
Q3 D
Q4 A
Q5 D
Q6 A
Q7 D
Q8 D
Q9 2
Q10 D
Q11 30
Q12 B
Q13 B
Q14 4
Q15 5

Q16 10
Q17 4
Q18 2
Q19 C
Q20 D
Q21 D
Q22 6
Q23 B
Q24 B
Q25 24
Q26 B
Q27 A
Q28 7
Q29 C
Q30 C



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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

$$(x - 2)(x + 3) > 0$$

$$x < -3 \text{ or } x > 2$$

It will be true for all natural numbers greater than 2.

Thus, there are infinitely many solutions.

Hence, option d is the correct answer.

Video Solution:



Q2 Text Solution:

$$2N + 5 > 3N$$

$$N < 5$$

So, the possible values of N are 1, 2, 3, and 4, i.e., four possible values.

Hence, option b is the correct answer.

Video Solution:



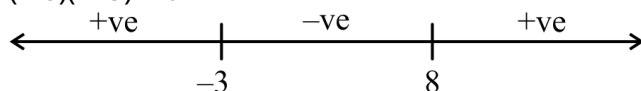
Q3 Text Solution:

$$(x-3)(x-2) > 30$$

$$x^2 - 5x + 6 > 30$$

$$x^2 - 5x - 24 > 0$$

$$(x-8)(x+3) > 0$$



$$\Rightarrow x < -3 \cup x > 8.$$

Video Solution:



Q4 Text Solution:

Given,

$$15x + 19 \leq 21x + 4$$

$$\text{or } 21x - 15x \geq 19 - 4$$

$$\text{or } 6x \geq 15$$

$$\text{or } x \geq \frac{15}{6}$$

$$\text{or } x \geq \frac{5}{2}$$

$$\text{And } 17x + 18 \geq 11x - 12$$

$$\text{or } 17x - 11x \geq -12 - 18$$

$$\text{or } 6x \geq -30$$

$$\text{or } x \geq -5$$

$$\text{Hence } x \text{ is } \geq \frac{5}{2} \text{ or } \left[\frac{5}{2}, \infty\right)$$

Hence, option a is the correct answer.

Video Solution:



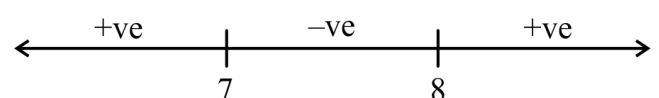
Q5 Text Solution:

$$x^2 - 15x + 56 \leq 0$$

$$x^2 - 8x - 7x + 56 \leq 0$$

$$x(x - 8) - 7(x - 8) \leq 0$$

$$(x - 8)(x - 7) \leq 0$$



$$7 \leq x \leq 8 \text{ or } [7, 8]$$

Hence, option d is the correct answer



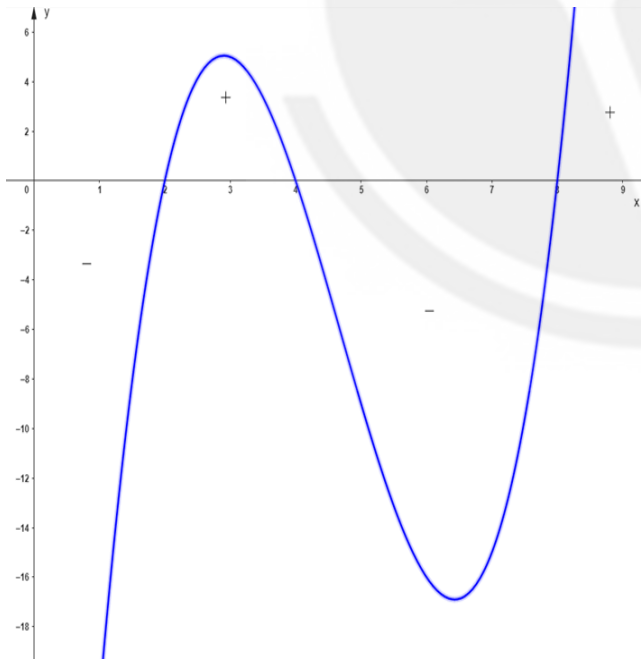
Video Solution:**Q6 Text Solution:**

For $(y - 3x)$ to be greatest, y should be maximum and x should be minimum

Here, maximum value of $y = 0$ and minimum value of $x = 2$

$$\text{So, } (y - 3x) = 0 - (3 \times 2) = -6$$

Ans. a

Video Solution:**Q7 Text Solution:**

From the above wavy curve, we can say that,
 $(x-2)(x-4)(x-8) \leq 0$

- $4 \leq x \leq 8$ or, $x \leq 2$

Also, x is a positive integer.

So, x can assume a minimum value of 1 and a maximum value of 8.

Therefore, $a = \text{minimum value of } x + 2 = 1 + 2 = 3$

$b = \text{maximum value of } x + 2 = 8 + 2 = 10$

Thus, $a + b = 3 + 10 = 13$

Video Solution:**Q8 Text Solution:**

$$3x + 5y < 7$$

$$\text{or, } x < \frac{7-5y}{3}$$

Since y can take only integral values, we can put $y = \text{any negative integer}$

by doing so, we can see that the value of x in the above inequality becomes larger and larger, as y keeps decreasing.

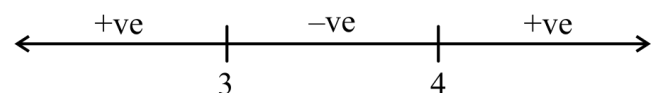
Thus, we cannot determine the maximum possible value of ' x '

Hence option d is correct answer

Video Solution:**Q9 Text Solution:****Topic - Inequalities**

$$x^2 - 7x + 12 \leq 0$$

$$\Rightarrow (x-4)(x-3) \leq 0$$



$$3 \leq x \leq 4$$

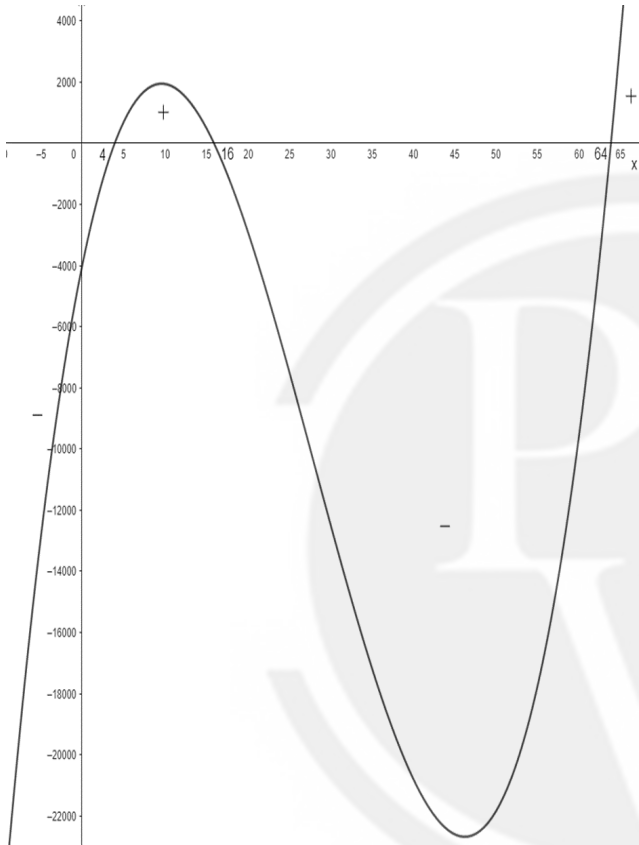
Hence x can take 2 integral values

Video Solution:



Q10 Text Solution:

The critical points for the first inequality will be 4, 16, and 64. Drawing them and checking for the values on the wavy curve



From the above wavy curve, we can say that if $16 < x < 64$ or $x < 4$, then the inequation is satisfied.

$$x^2 - 3x - 18 \leq 0$$

$$(x-6)(x+3) \leq 0$$

$$-3 \leq x \leq 6$$

Combining both the inequalities

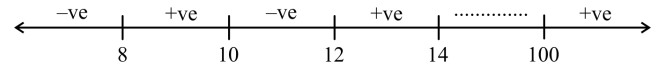
So, $x = -3, -2, -1, 0, 1, 2, 3$.

Thus, x can assume 7 integral values.

Video Solution:



Q11 Text Solution:



As can be seen from the above diagram, total number of critical points here will be the number of multiples of 2 from 8 to 100. From 1 to 100, we have 50 multiples of 2 but the first 3 multiples i.e. 2, 4 and 6 need to be excluded, therefore we have 47 critical points in all.

Thus, the total number of regions that the number line gets divided into will be $47 + 1 = 48$.

From the same, we can conclude, that there will be a total of 24 positive and 24 negative regions.

Out of the 23 negative regions that will lie to the right of 8, we will have 1 positive integral value from each of those regions, thus a total of 23 integer values here.

Lastly, the 24th negative region that is $x < 8$ will give us 7 more positive integer values from 1 to 7.

Thus, the required number of values will be $23 + 7 = 30$.

Video Solution:



Q12 Text Solution:

$$(x^2 - 2x - 8)(x^2 - x - 12) \leq 0$$



- $(x^2 - 4x + 2x - 8)(x^2 - 4x + 3x - 12) \leq 0$
- $(x-4)(x+2)(x-4)(x+3) \leq 0$
- $(x-4)^2(x+2)(x+3) \leq 0$

as, x is a real number, $(x-4)^2 \geq 0$

So, $(x+2)(x+3) \leq 0$

$-3 \leq x \leq -2$ and $x = 4$

Thus, we see that $x = 4$ is the only natural number value of ' x ' that is possible.

Video Solution:



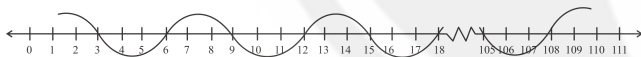
Q13 Text Solution:

Total number of terms here =

$$\frac{108-3}{3} + 1 = 36$$

So for 36 values here, the LHS=0

Lets mark deflection points on the number line.



A cycle of 3 is repeating itself in the graph, and 2 values between the cycle keep on alternating between the less than 0 and more than 0. (e.g., cycle 3-6, for 4 and 5, LHS<0, cycle 6-9, for 7,8, LHS>0, cycle 9-12, for 10,11, LHS<0, cycle 12-15, for 13,14, LHS>0, and so on)

There will be total = $\frac{108-3}{3} = 35$ cycles

Out of which 18 are satisfying the inequality. So integral values for which LHS<0 = $2 \times 18 = 36$

So number of integral values of x satisfying the inequality are = $36+36=72$

Video Solution:



Q14 Text Solution:

The given inequality is:

$$x^2 - 11x + 24 \leq 0$$

$$(x-3)(x-8) \leq 0$$

Using the Wavy – Curvy Method, we get



$$x \in [3, 8] \dots (i)$$

Also,

$$3 \geq \frac{7x+24}{4x+1}$$

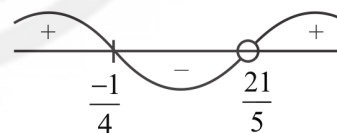
$$\frac{7x+24}{4x+1} - 3 \leq 0$$

$$\frac{7x+24-12x-3}{4x+1} \leq 0$$

$$\frac{-5x+21}{4x+1} \leq 0$$

$$\frac{x - \frac{21}{5}}{x + \frac{1}{4}} \geq 0$$

Using the Wavy-Curvy Method, we get



$$x \in \left(-\infty, -\frac{1}{4}\right) \cup \left[\frac{21}{5}, \infty\right) \dots (ii)$$

Therefore combining (i) and (ii), we get,

$$x \in \left[\frac{21}{5}, 8\right]$$

So, the 4 possible positive integer values of x are: 5, 6, 7 and 8.

Video Solution:




Q15 Text Solution:

The given inequality is:

$$\frac{1}{x+8} \leq \frac{1}{3x-5}$$

$$\frac{1}{x+8} - \frac{1}{3x-5} \leq 0$$

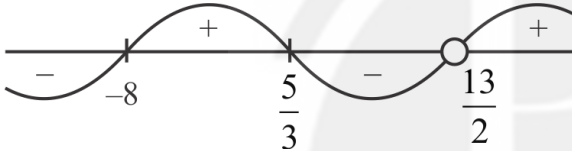
$$\frac{3x-5-x-8}{(x+8)(3x-5)} \leq 0$$

$$\frac{2x-13}{(x+8)(3x-5)} \leq 0$$

$$x - \frac{13}{2} \leq 0$$

$$(x+8)\left(x - \frac{13}{2}\right) \leq 0$$

Now, by using the Wavy-Curve Method, we get



i.e., $x \in (-\infty, -8) \cup \left(\frac{13}{2}, \infty\right)$

So, the number of positive integer values x can have, is 5, which are $x = 2, 3, 4, 5$ and 6 .

Video Solution:

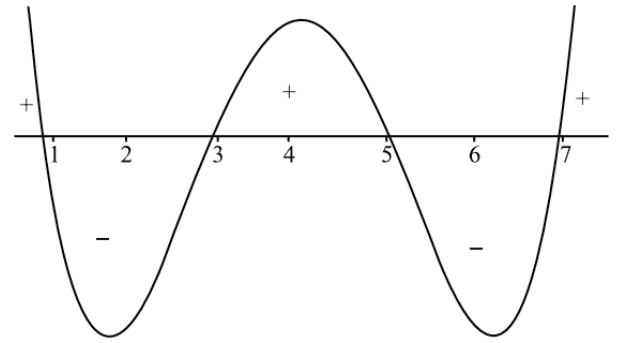
Q16 Text Solution:

As $(x-1)^{2016} \geq 0$, $(x-3)^{2018} \geq 0$, $(x-5)^{2020} \geq 0$ and

$(x-7)^{2022} \geq 0$, so

$$(x-1)^{2017}(x-3)^{2019}(x-5)^{2021}(x-7)^{2023} < 0$$

$$(x-1)(x-3)(x-5)(x-7) < 0$$



So, $1 < x < 3$; $5 < x < 7$

$$3 < 3x < 9; 15 < 3x < 21$$

So, $3x$ can assume 10 integral values.

Video Solution:

Q17 Text Solution:

The given inequality is:

$$\frac{x^2-6}{2x+7} \leq 3$$

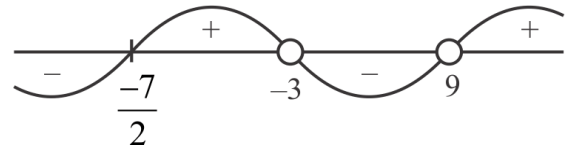
$$\frac{x^2-6}{2x+7} - 3 \leq 0 \quad \left[\begin{array}{l} \text{Since, we don't know} \\ \text{the sign of } 2x+7, \text{ we can't cross} \\ \text{multiply.} \end{array} \right]$$

$$\frac{x^2-6-6x-21}{2x+7} \leq 0$$

$$\frac{x^2-6x-27}{2x+7} \leq 0$$

$$\frac{(x-9)(x+3)}{x+\frac{7}{2}} \leq 0$$

Using the Wavy-Curve Method, we get



$$x \in \left(-\infty, -\frac{7}{2}\right) \cup [-3, 9]$$



So, x can have these 4 prime number values: 2, 3, 5, and 7.

Video Solution:



Q18 Text Solution:

The given inequality is:

$$\frac{x^3 - 8x^2 + 20x - 16}{(x^2 - x - 30)} \leq 0$$

$$\frac{x^2(x-2) - 6x(x-2) + 8(x-2)}{(x^2 - x - 30)} \leq 0$$

$$\frac{(x-2)(x^2 - 6x + 8)}{(x^2 - x - 30)} \leq 0$$

$$\frac{(x-2)(x^2 - 4x - 2x + 8)}{(x^2 - x - 30)} \leq 0$$

$$\frac{(x-2)^2(x-4)}{(x^2 - x - 30)} \leq 0$$

Therefore,

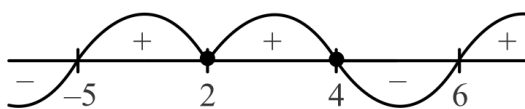
$$\frac{(x-2)^2(x-4)}{(x^2 - 6x + 5x - 30)} \leq 0$$

$$\frac{(x-2)^2(x-4)}{(x-6)(x+5)} \leq 0$$

Now, $x \neq -5, 6$, otherwise it will not be defined.

Also, x can be equal to 2 and 4 as equality holds.

Then, let's use the Wavy-Curvy method to find the ranges of values that x will attain.



● → Attainable

| → Non-Attainable

Hence, the required solution of the inequality is:

$$x \in (-\infty, -5) \cup \{2\} \cup [4, 6)$$

Thus, the smallest natural number ' x ' satisfying the given inequality is 2.

Video Solution:



Q19 Text Solution:

Here it is given that, $\frac{(x-10)(x-30)(x-50)}{(x-20)(x-40)(x-60)} \leq 0$

this can be written as,

$$\frac{(x-10)(x-30)(x-50)(x-20)(x-40)(x-60)}{(x-20)(x-40)(x-60)(x-20)(x-40)(x-60)} \leq 0$$

Critical points are: 10, 20, 30, 40, 50, 60.

Possible solutions lie between $[10, 20]$, $[30, 40]$, $[50, 60]$ = 33 solutions

But at $x = 20, 40, 60$ denominators become zero which is not allowed.

The possible number of solutions are $33 - 3 = 30$ solutions.

Video Solution:



Q20 Text Solution:

The given inequality is:

$$x^3 - 37x - 84 \geq 0$$

To solve this, first we need to factorize the given cubic polynomial. So, using the Trial and Error method, let's try to find a zero for it.

Look, at $x = -3$, the $x^3 - 37x - 84 = 0$. So, we can write it as:

$$x^2(x+3) - 3x(x+3) - 28(x+3) \geq 0$$

$$(x+3)(x^2 - 3x - 28) \geq 0$$

$$(x+3)(x^2 - 7x + 4x - 28) \geq 0$$

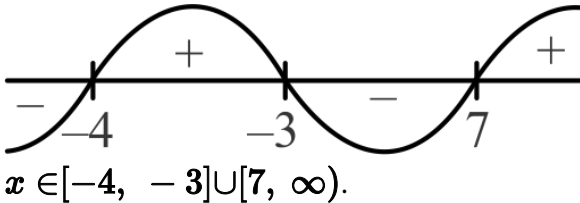
$$(x+3)[x(x-7) + 4(x-7)] \geq 0$$

$$(x+3)(x+4)(x-7) \geq 0$$

Here, the value x can be $-4, -3$ and 7 since equality holds.

So, let's use the Wavy-Curvy Method, we get





Video Solution:



Q21 Text Solution:

Let $y^3 = x$, then $y^6 - 37y^3 - 1728$ becomes $x^2 - 37x - 1728 \leq 0$

$$x^2 - 64x + 27x - 1728 \leq 0$$

$$x(x - 64) + 27(x - 64) \leq 0$$

The range of x will be $[-27, 64]$

Then, the range of y is $[-3, 4]$

Hence, option d is the correct answer

Video Solution:



Q22 Text Solution:

The given inequality can be written as:

$$\frac{(x-2)^{10}(x-8)^{12}(x-2)(x-8)}{(x-10)^8(x-6)^6(x-10)(x-6)} \leq 0$$

since,

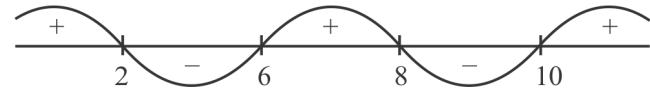
$$(x-2)^{10} \geq 0, (x-8)^{12} \geq 0, (x-10)^8 \geq 0, \text{ and } (x-6)^6 \geq 0$$

so we can rewrite the given inequality without them as:

$$\frac{(x-2)(x-8)}{(x-10)(x-6)} \leq 0$$

Now, the critical points are 2, 6, 8 and 10. But, x cannot be equal to 10 or 6, otherwise the inequality will fail to exist.

Now, let's solve this with the Wavy-Curve Method:



$$x \in [2, 6) \cup [8, 10)$$

i.e., x can have these 6 integer values: 2, 3, 4, 5, 8, and 9.

Video Solution:



Q23 Text Solution:

$$\frac{6x^2 - 25x + 11}{x^2 + 4x + 5} < 0$$

Now the denominator $x^2 + 4x + 5$ is always greater than zero for any value of x .

It means the numerator should be less than zero.

$$6x^2 - 25x + 11 < 0$$

$$6x^2 - 22x - 3x + 11 < 0$$

$$\Rightarrow (2x - 1)(3x - 11) < 0$$

$$\Rightarrow \frac{1}{2} < x < \frac{11}{3}$$

The only integers satisfying the above inequality are 1, 2, and 3.

So, required Sum = $(1 + 2 + 3) = 6$.

Video Solution:



Q24 Text Solution:

The given inequality can be written as



$$\frac{x^3 - 14x^2 + 59x - 70}{(x-1)(x-4)^2} \leq 0$$

$$\frac{x^2(x-2) - 12x(x-2) + 35(x-2)}{(x-1)(x-4)} \leq 0$$

$$\frac{(x^2 - 12x + 35)(x-2)}{(x-1)(x-4)} \leq 0$$

$$\frac{(x^2 - 7x - 5x + 35)(x-2)}{(x-1)(x-4)^2} \leq 0$$

$$\frac{[x(x-7) - 5(x-7)](x-2)}{(x-1)(x-4)^2} \leq 0$$

$$\frac{(x-2)(x-5)(x-7)}{(x-1)(x-4)^2} \leq 0$$

x neither can be 1, nor be 4, otherwise, the expression will become invalid.

Now, let's find out the range of x with the Wavy-Curve Method:



Hence, $x \in (1, 2] \cup [5, 7]$

So, the number of possible integer values of x are 2, 5, 6, 7.

i.e., we can have 4 integer values satisfying the given inequation.

Video Solution:



Q25 Text Solution:

The given inequality is:

$$\frac{x^2 - 6x - 26}{x^2 - 8x + 15} \leq \frac{x+9}{x-3} + \frac{1}{x-5}$$

$$\frac{x^2 - 6x - 26}{(x-3)(x-5)} \leq \frac{x+9}{x-3} + \frac{1}{x-5}$$

$$\frac{x^2 - 6x - 26}{(x-3)(x-5)} \leq \frac{(x+9)(x-5) + x-3}{(x-3)(x-5)}$$

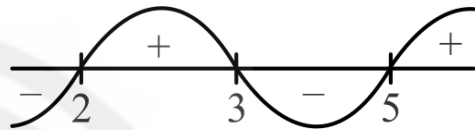
$$\frac{x^2 - 6x - 26}{(x-3)(x-5)} - \frac{x^2 + 4x - 45 + x - 3}{(x-3)(x-5)} \leq 0$$

$$\frac{x^2 - 6x - 26 - x^2 - 5x + 48}{(x-3)(x-5)} \leq 0$$

$$\frac{-11x + 22}{(x-3)(x-5)} \leq 0$$

$$\frac{(x-2)}{(x-3)(x-5)} \geq 0$$

Now, using the Wavy-Curve Method, we get



$x \in [2, 3] \cup (5, \infty)$

So, within the interval $x \in [0, 29]$, the possible integer values of x are: 2, 6, 7, 8, 9, ..., 28.

i.e., x can have 24 integer values within the given range.

Video Solution:



Q26 Text Solution:

The given inequality is:

$$\frac{7}{5} < \frac{x}{x+4} < \frac{9}{5}$$

$$7 < \frac{5x}{x+4} < 9 \dots (i)$$

$$\text{i.e., } 7 < \frac{5x}{x+4} \dots (ii) \text{ and } \frac{5x}{x+4} < 9 \dots (iii)$$

From (ii), we get

$$\frac{5x}{x+4} - 7 > 0 \quad [\text{Since, we don't know the sign of } x + 4, \text{ we cannot cross-multiply.}]$$

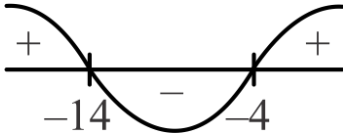


$$\frac{5x-7x-28}{x+4} > 0$$

$$\frac{-2x-28}{x+4} > 0$$

$$\frac{x+14}{x+4} < 0$$

Using the Wavy-Curve Method, we get



$$x \in (-14, -4) \dots (iv)$$

Now, from (iii), we get

$$\frac{5x}{x+4} < 9$$

$$\frac{5x}{x+4} - 9 < 0$$

$$\frac{5x-9x-36}{x+4} < 0$$

$$\frac{-4x-36}{x+4} < 0$$

$$\frac{x+9}{x+4} > 0$$

So, using the Wavy-Curve Method, we get



$$x \in (-\infty, -9) \cup (-4, \infty) \dots (v)$$

Combining (iv) and (v), we get

$$-14 < x < -9.$$

i.e., $x = -13, -12, -11$, and -10 .

So, x can take 4 integer values.

Video Solution:



Q27 Text Solution:

$$\text{Let } y = \frac{-x^2 + 6x - 1}{x}$$

$$xy = -x^2 + 6x - 1$$

$$x^2 + (y-6)x + 1 = 0$$

If x is real then,

$$(y-6)^2 - 4 \geq 0$$

$$y^2 - 12y + 36 - 4 \geq 0$$

$$y^2 - 12y + 32 \geq 0$$

$$(y-4)(y-8) \geq 0$$

$$y \leq 4 \text{ or } y \geq 8$$

$$a = 4 \text{ and } b = 8$$

Thus, the required option (A) is correct.

Video Solution:



Q28 Text Solution:

The given inequalities are:

$$(5+x)^4(2-x)^3 < 0 \dots (i) \text{ and}$$

$$-(5-x)^3(9+x)^5 < 0 \dots (ii)$$

From (i), we get

$$(5+x)^4(2-x)^3 < 0$$

$(5+x)^4$ is always positive, since, if $x = -5$, then equality will hold and that will be a contradiction.

So, we can write equation (i) as:

$$(2-x)^3 < 0$$

$$\Rightarrow (x-2)^3 > 0$$

$$\Rightarrow x > 2 \dots (iii)$$

Now, from (ii), we get

$$-(5-x)^3(9+x)^5 < 0$$

$$\Rightarrow (x-5)^3(x+9)^5 < 0$$

$$\Rightarrow (x-5)^2(x-5)(x+9)^4(x+9) < 0$$

Since, $(x-5)^2 \geq 0$, and $(x+9)^4 \geq 0$, so we can write

$$(x-5)(x+9) < 0$$

So, by using the Wavy-Curve Method, we have



$$x \in (-9, 5) \dots (iv)$$



Thus, combining (iii) and (iv), we have

$$x = 3, 4$$

So, the sum of these values will be $3 + 4 = 7$

Video Solution:



Q29 Text Solution:

The given inequality is:

$$\frac{1}{x^2+x-12} - \frac{1}{x^2+11x+28} \leq 0$$

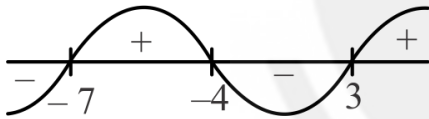
$$\frac{1}{(x-3)(x+4)} - \frac{1}{(x+4)(x+7)} \leq 0$$

$$\frac{x+7-x+3}{(x-3)(x+4)(x+7)} \leq 0$$

$$\frac{10}{(x-3)(x+4)(x+7)} \leq 0$$

The critical points for the inequality are: -7, -4 and 3. But, x cannot be equal to -7 or -4 or 3, otherwise the inequality will fail to exist.

Now, using the Wavy-Curvy Method, we get



$$x \in (-\infty, -7) \cup (-4, 3) \dots (i)$$

Also, it is given that, $-9 \leq x \leq 9 \dots (ii)$

Combining (i) and (ii), we get

$x = -9, -8, -3, -2, -1, 0, 1, \text{ and } 2.$

i.e., the required absolute difference = $|-9 - 2| = 11$

Video Solution:



Q30 Text Solution:

Here we have been given

$$3 \leq \frac{x+4}{2x+1} \leq 6$$

$$\frac{x+4}{2x+1} \geq 3$$

$$\frac{x+4}{2x+1} - 3 \geq 0$$

$$\frac{1-5x}{2x+1} \geq 0$$

Thus,

$$\frac{-1}{2} < x \leq \frac{1}{5} \quad (1)$$

$$\frac{x+4}{2x+1} \leq 6$$

$$\frac{x+4}{2x+1} - 6 \leq 0$$

$$\frac{-(11x+2)}{2x+1} \leq 0$$

$$\frac{11x+2}{2x+1} \geq 0$$

$$x < \frac{-1}{2} \text{ or } x \geq \frac{-2}{11} \quad (2)$$

Combining (1) and (2), we see that

$$\frac{-2}{11} \leq x \leq \frac{1}{5}$$

On multiplication with 55,

$$-10 \leq 55x \leq 11$$

Therefore, $55x$ can assume all the values from -10 to 11.

Thus, the integral numbers that can be assumed by $55x$ will be 22.

Video Solution:



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